HARE.jl — Ecosystem Comparison

This document compares HARE.jl against existing implementations of the same estimation methods across open-source and commercial platforms. The prose discussion focuses on open-source packages (R and Python); Stata and SAS are included in the comparison tables for completeness but are excluded from the open-source narrative.

All HARE methods are textbook-standard procedures covered in Greene (2018), *Econometric Analysis*, 8th edition. Relevant sections are cited for each method.

Search note. Comparisons for heteroscedasticity correction were conducted using both author-name searches (Harvey, Glejser) and link-function terminology (exponential, quadratic, linear variance link, FGLS heteroscedasticity). The closest open-source analog to HARE's heteroskedasticity estimators is `nlme::gls()` with variance functions (`varFunc`): `varExp()` implements the exponential variance model (§1.1), `varPower()` the power variance model (§1.2) — and with δ fixed at 1, approximates the quadratic link (§1.3) — and `varIdent()` groupwise heteroskedasticity (§1.6). All three are estimated by ML or REML — `nlme` does not offer two-step or iterated FWLS.

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1. Heteroscedasticity

1.1 Exponential variance link — Harvey (1976), `LINK=EXP`

Greene §9.7.1 "Multiplicative Heteroscedasticity"

Model: $\sigma_i^2 = \exp(\gamma_0 + \mathbf{z}_i^\top \gamma)$

	HARE.jl	R	Python	Stata	SAS
Two-step FWLS	yes `exponential_fwls`	—	—	yes `hetregress, twostep`	—
Iterated FWLS	yes `exponential_ifwls`	—	—	—	—
MLE	yes `exponential_mle`	partial `nlme::gls`	—	yes `hetregress`	yes `HETERO LINK=EXP`
Harvey test	yes `harvey_test`	yes `skedastic`	—	—	yes `HETERO TEST=LM`

R `nlme::gls(weights = varExp(form = ~ z))` (base R recommended package) models $\mathrm{Var}(\varepsilon_i) = \sigma^2 \exp(2\delta z_i)$, which is Harvey's exponential variance model with $\gamma_0 = \log \sigma^2$ and $\gamma = 2\delta$. For multiple auxiliary regressors, `varExp(form = ~ z1 + z2)` estimates a δ_j per regressor, giving $\log \sigma_i^2 = \log \sigma^2 + 2\sum_j \delta_j z_{ij}$ — equivalent to Harvey up to this factor-of-2 reparameterisation. Estimation is by ML or REML jointly over (β, δ), so the nlme row is comparable to HARE's `exponential_mle`, not to the two-step or iterated FWLS variants (`exponential_fwls`, `exponential_ifwls`). No FWLS path exists in nlme.

R `skedastic` (CRAN, active) implements Harvey (1976) as a *test* only — it computes the test statistic and p-value but produces no corrected beta or consistent gamma estimates.

Python: No implementation found under either the Harvey name or exponential-link terminology.

Stata: `hetregress` implements the exponential variance link as both a two-step GLS estimator and as MLE. The two-step variant is directly comparable to HARE's `exponential_fwls`; an iterated version is not documented.

SAS: `PROC AUTOREG` with the `HETERO` statement and `LINK=EXP` implements the exponential variance model by **MLE only** — two-step FGLS is not offered. SAS maximises the normal log-likelihood jointly over β and γ, which is asymptotically efficient under correct specification. The `TEST=LM` option produces the Lagrange multiplier test for heteroscedasticity of the specified form. SAS allows arbitrary auxiliary regressors $\mathbf{z}_i$; HARE matches this generality via the optional `Z` argument (defaulting to the augmented mean regressors).

HARE implementation notes. The FWLS auxiliary regression subtracts `c = E[log(χ²(1))] = digamma(1/2) + log(2) = -1.2703628454614782` from `log(u_hat²)` before fitting, centring the error term and yielding consistent estimates of *all* gamma coefficients regardless of regressor means. Without this correction (which is not documented in Stata or SAS) the intercept and all slope gammas are biased when regressors have non-zero means. The MLE variant (`exponential_mle`) jointly maximises the normal log-likelihood over (β, γ) via L-BFGS and is asymptotically efficient.

1.2 Power variance link

Model: $\sigma_i^2 = \sigma^2 |\mathbf{z}_i|^{2\delta}$, equivalently $\log \sigma_i^2 = \gamma_0 + \sum_j \gamma_j \log |z_{ij}|$ with $\gamma_0 = \log \sigma^2$, $\gamma_j = 2\delta_j$

	HARE.jl	R	Python	Stata	SAS
Two-step FWLS	yes `power_fwls`	—	—	—	—
Iterated FWLS	yes `power_ifwls`	—	—	—	—
MLE	yes `power_mle`	partial `nlme::gls`	—	—	—

R `nlme::gls(weights = varPower(form = ~ abs(z)))` models $\mathrm{Var}(\varepsilon_i) = \sigma^2 |z_i|^{2\delta}$, which is the power variance model with $\gamma_0 = \log \sigma^2$ and $\gamma_1 = 2\delta$. This is exactly the HARE power model. When `varPower` estimates δ from data it maximises the same log-likelihood as `power_mle`, so the two are numerically equivalent (cross-validated in §7.2).

HARE implementation. `power_fwls` and `power_ifwls` work by applying the exponential FWLS procedure (Harvey bias correction) on $\log |Z|$ in place of Z, yielding exact OLS estimates of $\gamma = (\log \sigma^2, 2\delta_1, ...)$ with exact standard errors — no delta method approximation is needed. `power_mle` pre-log-transforms Z and then calls the same exponential MLE kernel; the reported parameters $(\gamma_0, \gamma_1, ...)$ are directly interpretable as $(\log \sigma^2, 2\delta_1, ...)$.

1.3 Quadratic variance link — Glejser (1969), `LINK=SQUARE`

Greene §9.5.3 "Estimation When Ω Contains Unknown Parameters"

Model: $\sigma_i = \gamma_0 + \mathbf{z}_i^\top \gamma$ (equivalently, σ_i^2 is quadratic in $\mathbf{z}_i$)

	HARE.jl	R	Python	Stata	SAS
Two-step FWLS	yes `quadratic_fwls`	—	—	—	—
Iterated FWLS	yes `quadratic_ifwls`	—	—	—	—
MLE	yes `quadratic_mle`	partial `nlme::gls`	—	—	yes `HETERO LINK=SQUARE`
Glejser test	yes `glejser_test`	yes `skedastic`	—	—	—

R `nlme::gls(weights = varPower(value = 1, form = ~ z, fixed = TRUE))` models $\mathrm{Var}(\varepsilon_i) \propto |z_i|^2$, i.e. $\sigma_i \propto |z_i|$. This is **not** Glejser's model, which is $\sigma_i = \gamma_0 + \mathbf{z}_i^\top \gamma$ (a shifted linear function of z_i). The power model forces

$\sigma_i = 0$ when $z_i = 0$ and has no intercept, whereas Glejser's model allows a non-zero baseline variance at $z_i = 0$. The two overlap empirically when z_i varies well away from zero and the intercept is small, but they are not equivalent in functional form. When δ is estimated from data (rather than fixed at 1), `varPower(form = ~ z)` fits the power exponent jointly with β by ML.

`skedastic` provides the Glejser test statistic only; no open-source package implements Glejser's linear-SD auxiliary regression as a correction estimator.

SAS `HETERO LINK=SQUARE` specifies $\sigma_i^2 = \sigma^2(1+\mathbf{z}_i^\top\eta)^2$, i.e. $\sigma_i = \sigma(1+\mathbf{z}_i^\top\eta)$. Absorbing the scale into the coefficients — $\gamma_0 = \sigma$, $\gamma = \sigma\eta$ — gives $\sigma_i = \gamma_0 + \mathbf{z}_i^\top\gamma$, which is exactly the Glejser functional form (since HARE's $\mathbf{z}_i$ always includes a constant column). The two models are therefore **equivalent in functional form**; they differ only in estimation: SAS uses **joint MLE** over (β, σ, η); HARE offers both FWLS (`quadratic_fwls`) and joint MLE (`quadratic_mle`).

HARE FWLS implementation note. Because `E[|e|] = sqrt(2/pi)` for e ~ N(0,1), OLS of `|u_hat|` on Z converges to `gamma × sqrt(2/pi)`. HARE corrects by multiplying the raw OLS coefficients by `sqrt(pi/2)` and the vcov by `pi/2`. Without this correction gamma is biased toward zero by the factor `sqrt(2/pi) ≈ 0.798` for all sample sizes.

1.4 Linear variance link — Breusch-Pagan (1979), `LINK=LINEAR`

No dedicated textbook estimator section; the test appears as the Breusch-Pagan test in Greene §9.5

Model: $\sigma_i^2 = \gamma_0 + \mathbf{z}_i^\top\gamma$ (variance linear in $\mathbf{z}_i$)

	HARE.jl	R	Python	Stata	SAS
MLE	yes `linear_mle`	—	—	—	yes `HETERO LINK=LINEAR`
Breusch-Pagan test	yes `HypothesisTests.jl`	yes `lmtest::bptest`	yes `statsmodels`	yes	yes

This is a distinct parameterisation from both the exponential and quadratic links — the *variance* (not the log-variance, not the standard deviation) is linear in $\mathbf{z}_i$. This is exactly the variance model underlying the **Breusch-Pagan (1979) test** for heteroskedasticity: the LM test for $H_0\colon \gamma = \mathbf{0}$ in this model. HypothesisTests.jl implements it as `BreuschPaganTest(X, e)`, documented as equivalent to `WhiteTest(X, e, type = :linear)`.

Note on `nlme::varPower(delta = 0.5)`. A power model with $\delta = 0.5$ gives $\mathrm{Var}(\varepsilon_i) \propto |z_i|$, i.e. variance linear in $|z_i|$. This is superficially similar to the Breusch-Pagan linear variance model but is not equivalent: it lacks an additive intercept and constrains the constant term to zero. The `varPower` family does not expose a `LINK=LINEAR`-style parameterisation.

No open-source *estimation* implementation (i.e. corrected beta) with the exact linear-variance functional form was found in R or Python. HARE's `linear_mle` jointly maximises the normal log-likelihood over (β, γ) via L-BFGS, matching SAS's approach.

1.5 Second-order Taylor expansion — White (1980)

Greene §9.5.2 "White's General Test"

Model: $\sigma_i^2 \approx \gamma_0 + \sum_j \gamma_j z_{ij} + \sum_j \gamma_{jj} z_{ij}^2 + \sum_{j<k} \gamma_{jk} z_{ij} z_{ik}$

The variance function is approximated by a second-order Taylor expansion in the auxiliary regressors $\mathbf{z}_i$ (typically $\mathbf{z}_i = \mathbf{x}_i$, the mean regressors). For p auxiliary regressors the variance equation has $1 + p(p+3)/2$

free parameters: one intercept, p linear terms, p squared terms, and $\binom{p}{2}$ cross-product terms. With $p = 1$: 3 gammas; with $p = 2$: 6 gammas; with $p = 3$: 9 gammas — growing quadratically in p.

	HARE.jl	R	Python	Stata	SAS
FWLS / MLE estimator	— *(planned)*	—	—	—	—
White test	yes `HypothesisTests.jl`	yes `lmtest::bptest`	yes `statsmodels`	yes	yes

The **White (1980) test** is the LM test for $H_0 \colon \gamma = \mathbf{0}$ in this variance model. HypothesisTests.jl implements it as `WhiteTest(X, e)` (also aliased as `BreuschPaganTest(X, e, type = :white)`). No open-source platform implements the second-order Taylor expansion as an *estimation* method (corrected beta); all provide the test statistic only.

HARE planned implementation. With $\mathbf{z}_i = \mathbf{x}_i$ (the default), the full expanded auxiliary regressor matrix can be constructed automatically. The natural estimator would be `quadratic_expansion_mle` (joint MLE over (β, γ)), analogous to `exponential_mle`. The large number of gamma parameters relative to the exponential link means this approach is most useful when heteroskedasticity is suspected but the functional form is unknown.

1.6 Groupwise heteroscedasticity — `two_step_groupwise`, `iterated_groupwise`, `groupwise_mle`

Greene §9.7.2 "Groupwise Heteroscedasticity"

Model: $\mathrm{Var}(\varepsilon_i) = \sigma_g^2$ for all i in group g (constant variance within groups, unrestricted across groups).

	HARE.jl	R	Python	Stata	SAS
Two-step FWLS	yes `two_step_groupwise`	—	—	—	yes
Iterated (= MLE at convergence)	yes `iterated_groupwise`	partial `nlme::gls`	—	—	yes
Explicit MLE	yes `groupwise_mle`	partial `nlme::gls`	—	—	yes

R `nlme::gls(weights = varIdent(form = ~1 | group))` estimates a separate scale parameter per group, producing group-specific standard deviations σ_g. The functional form is identical to HARE's groupwise model. Estimation is by ML or REML; nlme does not offer a FWLS path. `groupwise_mle` is cross-validated against `nlme::gls(varIdent)` in §7.2.

No Python implementation found. SAS `PROC AUTOREG` supports groupwise heteroscedasticity via the `HETERO` statement with group dummy variables.

Estimation. The MLE first-order condition for σ_g^2 is exactly the within-group sample variance of residuals: $\hat{\sigma}_g^2 = \frac{1}{n_g} \sum_{i \in g} \hat{u}_i^2$. The iterated FWLS therefore converges to the joint MLE of $(\beta, \sigma_1^2, \ldots, \sigma_G^2)$ at convergence. `groupwise_mle` runs the same iteration internally and is the recommended call when the primary goal is the MLE estimate.

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2. AR(1) Autocorrelation

2.1 Cochrane-Orcutt — AR(1) GLS dropping the first observation

Greene §20.9.1 "AR(1) Disturbances"

	HARE.jl	R	Python	Stata	SAS
Two-step	yes `two_step_cochrane_orcutt`	yes `prais`	yes `GLSAR`	yes `prais, corc`	yes (fallback)
Iterated	yes `iterated_cochrane_orcutt`	yes `prais`	yes `GLSAR`	yes `prais, corc`	yes (fallback)

Cochrane-Orcutt quasi-differences the data starting from observation 2, discarding the first observation and losing one degree of freedom compared with Prais-Winsten. **Prais-Winsten strictly dominates Cochrane-Orcutt**: it retains the first observation via the scaling `sqrt(1 - rho^2)` at zero additional cost, yielding coefficient estimates with smaller variance. HARE provides `two_step_cochrane_orcutt` and `iterated_cochrane_orcutt` for compatibility with software that uses the CO convention. `statsmodels.GLSAR` (Python) always uses the Cochrane-Orcutt convention (first observation dropped).

2.2 Prais-Winsten — AR(1) GLS

Greene §20.9.1 "AR(1) Disturbances"

	HARE.jl	R	Python	Stata	SAS
Two-step	yes `two_step_prais_winsten`	yes `prais`	—	yes `prais`	yes `METHOD=YW`
Iterated	yes `iterated_prais_winsten`	yes `prais`	—	yes `prais`	yes `METHOD=ITYW`

R `prais` (CRAN, active, July 2025): closest open-source counterpart. If estimated rho exceeds 1 during an iteration, the package falls back silently to Cochrane-Orcutt (drops the first observation). Also supports panel data.

Stata `prais`: implements both Prais-Winsten (default) and Cochrane-Orcutt (`corc` option), two-step and iterated.

SAS `PROC AUTOREG METHOD=YW`: the Yule-Walker method is equivalent to Prais-Winsten for AR(1). `METHOD=ITYW` is the iterated version. SAS documentation refers to these as "two-step full transform" following Harvey (1981).

HARE always applies the Prais-Winsten scaling `sqrt(max(1 - rho², 0))`, avoiding the imaginary-value edge case without silently changing methods.

2.3 Hildreth-Lu — grid search for AR(1) rho

Greene §20.9.1 "AR(1) Disturbances"

	HARE.jl	R	Python	Stata	SAS
Grid search	yes `hildreth_lu`	partial `HoRM`	—	—	yes `METHOD=HL`

R `HoRM` (CRAN, actively maintained — v0.1.4, May 2025; supplement to Young's *Handbook of Regression Methods*): exports `hildreth.lu(y, x, rho)`, which evaluates the Cochrane-Orcutt-transformed regression for a **single fixed rho**. This is a building block, not a complete estimator — the user must write the grid search loop manually to minimise SSE over a rho grid. HARE's `hildreth_lu` wraps this loop internally (200-point uniform grid on $(-0.99, 0.99)$) and returns the best-fitting model in one call.

SAS `PROC AUTOREG METHOD=HL` implements the same full grid search and uses the same Cochrane-Orcutt convention (first observation dropped).

2.4 Beach-MacKinnon (1978) — exact MLE for AR(1)

Greene §20.9.1 "AR(1) Disturbances"

	HARE.jl	R	Python	Stata	SAS
Exact MLE	yes `beach_mackinnon`	partial `nlme::gls`	—	—	yes `METHOD=ML`

R `nlme::gls(corAR1(), method="ML")` (base R recommended package, always available): maximises the exact normal log-likelihood for a GLS model with an AR(1) correlation structure, including the Jacobian contribution of the first observation — which is precisely what Beach & MacKinnon (1978) derive. It is therefore a **functional equivalent**, though it is not named after Beach-MacKinnon and lives in a broader mixed-effects framework. Cross-validation on our reference dataset (see Section 8) confirms agreement to $\sim 10^{-8}$ in ρ and $\sim 10^{-9}$ in coefficients — differences attributable to different numerical optimisers converging to the same maximum.

HARE implements the concentrated likelihood (beta profiled out analytically) and maximises over rho alone using Brent's method. `nlme::gls` optimises jointly over all parameters, which is asymptotically equivalent and numerically indistinguishable in practice.

SAS `PROC AUTOREG METHOD=ML` cites Beach & MacKinnon (1978) explicitly.

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3. Heteroscedasticity and AR(1) Autocorrelation

3.1 Sequential HARE — AR(1) + exponential heteroskedasticity (sequential)

Greene §9.7.1 + §20.9.1

	HARE.jl	R	Python	Stata	SAS
Two-step	yes `two_step_sequential`	—	—	—	—
Iterated	yes `iterated_sequential`	—	—	—	—

Not available on any other platform. SAS `PROC AUTOREG` explicitly prohibits combining the `HETERO` statement (heteroscedasticity) with the `NLAG=` option (AR(1)) unless GARCH is also specified. Stata's `prais` and `hetregress` are separate commands with no combined mode.

3.2 Joint HARE — AR(1) + exponential heteroskedasticity (joint MLE)

Greene §9.7.1 + §20.9.1

	HARE.jl	R	Python	Stata	SAS
Two-step (concentrated MLE)	yes `two_step_joint`	—	—	—	—
Iterated (coordinate descent)	yes `iterated_joint`	—	—	—	—

Not available on any other platform, open-source or commercial.

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4. Summary gap tables

4.1 What HARE has that no other open-source platform has

Method	Greene reference	Stata	SAS
Exponential variance link FWLS two-step and iterated	§9.7.1	two-step only	—
Exponential variance link MLE	§9.7.1	yes `hetregress`	yes `LINK=EXP`
Exponential variance link LM test (`harvey_test`)	§9.7.1	—	yes `TEST=LM`
Power variance link FWLS two-step and iterated	—	—	—
Power variance link MLE	—	—	—
Quadratic variance link FWLS two-step and iterated	§9.5.3	—	—
Quadratic variance link MLE	§9.5.3	—	yes `LINK=SQUARE`
Quadratic variance link LM test (`glejser_test`)	§9.5.3	—	—
Linear variance link MLE	—	—	yes `LINK=LINEAR`
Prais-Winsten (retains first observation)	§20.9.1	yes	yes
Cochrane-Orcutt (drops first observation)	§20.9.1	yes	yes (fallback)
Hildreth-Lu grid search	§20.9.1	—	yes
Beach-MacKinnon exact MLE	§20.9.1	—	yes
Groupwise heteroscedasticity FWLS, iterated, and MLE	§9.7.2	—	yes
Sequential AR(1) + exponential heteroskedasticity	§9.7.1 + §20.9.1	—	—
Joint AR(1) + exponential heteroskedasticity MLE	§9.7.1 + §20.9.1	—	—

4.2 What other platforms have that HARE does not (yet)

Functionality	Platform	Notes
Panel data AR(1) correction	R `prais`, Stata `xtgls`	HARE is cross-section / time series only
AR(1) + Glejser/quadratic combined estimator	—	Sequential and Joint HARE use the exponential link only

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5. Implementation quality comparison

Detail	HARE	R `prais`	Python `GLSAR`	Stata `prais`	SAS `METHOD=YW`
First observation	PW: retained; CO: dropped	Retained; falls back if rho > 1	Dropped (CO)	Retained (default); CO via `corc`	Retained
Harvey bias correction	Subtracts c from LHS; all gammas consistent	N/A	N/A	Not documented	Not documented
Glejser sqrt(pi/2) correction	yes	N/A	N/A	N/A	N/A
Exact AR(1) likelihood	yes `beach_mackinnon`	partial `nlme::gls`	—	—	yes `METHOD=ML`
Combined AR(1) + heteroscedasticity	yes	—	—	—	—
Panel support	—	yes	—	yes	yes
Iterated exponential FWLS	yes	N/A	N/A	—	N/A
Open-source MIT licence	yes	yes	yes	—	—

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6. Key finding

The combined AR(1) + exponential heteroskedasticity estimators — both sequential and joint MLE — are **not available on any other platform, open-source or commercial**. For the individual methods, the closest commercial coverage is SAS `PROC AUTOREG`, which implements Hildreth-Lu, Beach-MacKinnon ML, and exponential variance MLE (but not exponential FWLS or the quadratic/power links), yet still cannot combine AR(1) correction with heteroscedasticity correction outside of GARCH models.

HARE.jl therefore brings to the open-source Julia ecosystem a complete and correct implementation of the classical FGLS correction methods for heteroscedasticity and AR(1) autocorrelation — methods taught in every graduate econometrics course (Greene §9.7.1, §20.9.1) and available in SAS for decades — with no equivalent open-source implementation in R, Python, or Julia prior to this package.

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7. Cross-Validating Against R

To validate numerical correctness independently of unit tests, HARE.jl results are compared against R reference implementations on an identical dataset. The data are generated once in Julia (`StableRNG(42)`, $n = 500$) and the exact same observations are passed to R via CSV export. No random number generation occurs in R; agreement between the two implementations is therefore a direct test of algorithmic equivalence, not just statistical consistency.

R session: R 4.5.1 · prais 1.1.4 · orcutt 2.3 · skedastic 2.0.3 · nlme 3.1.168.

The cross-validation tests live in `test/runtests_r.jl` and require no R installation to run — the reference values below are hardcoded.

7.1 Software correspondence and test formulas

HARE.jl function	R package :: function	Formula / method
`iterated_prais_winsten`	`prais::prais_winsten()`	Iterates $\hat{\rho} = \frac{\sum \hat{u}_t \hat{u}_{t-1}}{\sum \hat{u}_{t-1}^2}$ + PW transform until $\lvert\hat{\rho}^{(i)} - \hat{\rho}^{(i-1)}\rvert < \text{tol}$
`two_step_prais_winsten`	`prais::prais_winsten(twostep=TRUE)`	Single pass: OLS → $\hat{\rho}$ → PW-GLS
`iterated_cochrane_orcutt`	`orcutt::cochrane.orcutt()`	Same iteration as PW but drops first observation
`beach_mackinnon`	`nlme::gls(corAR1(), method="ML")`	Exact AR(1) log-likelihood maximised over (β, ρ); first observation included
`exponential_mle`	`nlme::gls(weights=varExp(form=~z), method="ML")`	Same likelihood; $\gamma_0 = \log \sigma^2_{\text{nlme}}$, $\gamma_1 = 2\delta_{\text{nlme}}$
`power_mle`	`nlme::gls(weights=varPower(form=~abs(z)), method="ML")`	Same likelihood; $\gamma_0 = \log \sigma^2_{\text{nlme}}$, $\gamma_1 = 2\delta_{\text{nlme}}$
`groupwise_mle`	`nlme::gls(weights=varIdent(form=~1|group), method="ML")`	Same likelihood; group σ_g^2 directly reported
`harvey_test`	`skedastic::harvey()`	LM = ESS / $\psi_1(\frac{1}{2})$, where $\psi_1(\frac{1}{2}) = \pi^2/2 = \text{Var}(\log \chi^2(1))$
`glejser_test`	`skedastic::glejser()`	LM = $\text{ESS}_{\lvert\hat{u}\rvert}$ / $(\hat{\sigma}^2(1 - 2/\pi))$, where $(1 - 2/\pi) = \text{Var}(\lvert N(0,1)\rvert)$

Not cross-validated (no direct R equivalent): `hildreth_lu` (HoRM evaluates only a fixed rho, not the full grid search), all FWLS estimators (`exponential_fwls`, `exponential_ifwls`, `quadratic_fwls`, `quadratic_ifwls`, `power_fwls`, `power_ifwls`), `two_step_sequential`, `iterated_sequential`, `two_step_joint`, `iterated_joint`.

7.2 Numerical results

AR(1) dataset — true parameters $\beta = [1, 2, -1]$, $\rho = 0.7$.

Quantity	HARE.jl	R	Max |diff|
`iterated_prais_winsten` $\hat{\beta}$	[0.807539, 2.000888, −0.995887]	[0.807539, 2.000888, −0.995887]	$< 10^{-10}$
`iterated_prais_winsten` $\hat{\rho}$	0.747516	0.747516	$< 10^{-12}$
`two_step_prais_winsten` $\hat{\beta}$	[0.807132, 2.000733, −0.995997]	[0.807132, 2.000733, −0.995997]	$< 10^{-12}$
`two_step_prais_winsten` $\hat{\rho}$	0.741971	0.741971	$< 10^{-12}$

Quantity	HARE.jl	R	Max |diff|
`iterated_cochrane_orcutt` $\hat{\beta}$	[0.804907, 2.000953, −0.995974]	[0.804907, 2.000953, −0.995974]	$< 10^{-10}$
`iterated_cochrane_orcutt` $\hat{\rho}$	0.747510	0.747510	$< 10^{-12}$

Heteroskedastic dataset — Harvey DGP, $\sigma_i = \exp(0.5\, x_{1i})$, auxiliary regressors $Z = [1, x_1, x_2]$ ($p = 2$ slope parameters).

Quantity	HARE.jl	R	|diff|
`harvey_test` LM (het DGP)	122.0117	122.0117	$< 10^{-10}$
`harvey_test` p-value (het DGP)	3.20×10^{-27}	3.20×10^{-27}	$< 10^{-32}$
`harvey_test` LM (hom DGP)	1.1127	1.1127	$< 10^{-10}$
`harvey_test` p-value (hom DGP)	0.5733	0.5733	$< 10^{-10}$
`glejser_test` LM (het DGP)	188.1612	188.1612	$< 10^{-10}$
`glejser_test` p-value (het DGP)	1.38×10^{-41}	1.38×10^{-41}	$< 10^{-46}$
`glejser_test` LM (hom DGP)	0.8859	0.8859	$< 10^{-10}$
`glejser_test` p-value (hom DGP)	0.6422	0.6422	$< 10^{-10}$

Differences at the level of 10^{-10}–10^{-12} reflect standard double-precision floating-point rounding in the two independent implementations (Julia and R). There are no algorithmic discrepancies.

MLE heteroskedastic datasets — true parameters $\beta = [1, 2, -1]$; exponential DGP: $\log \sigma_i^2 = x_{1i}$; power DGP: $\sigma_i^2 = |x_{1i}|$; groupwise DGP: $\sigma_g \in \{0.5, 1.0, 1.5, 2.0\}$, $G = 4$ equal groups; all with $n = 500$, `StableRNG(99)`. Data generated independently of the AR(1)/test datasets (`StableRNG(42)`).

Quantity	HARE.jl	R (`nlme`)	Max |diff|
`exponential_mle` $\hat{\beta}$	[0.9763, 1.9934, −1.0115]	[0.9763, 1.9934, −1.0115]	$< 10^{-4}$
`exponential_mle` $\hat{\gamma} = (\log \hat{\sigma}^2, 2\hat{\delta})$	[0.0039, 1.0801]	[0.0039, 1.0801]	$< 10^{-4}$
`exponential_mle` log-likelihood	−694.669	−694.669	$< 10^{-3}$
`power_mle` $\hat{\beta}$	[0.9955, 2.1150, −0.9639]	[0.9955, 2.1150, −0.9639]	$< 10^{-4}$
`power_mle` $\hat{\gamma} = (\log \hat{\sigma}^2, 2\hat{\delta})$	[−0.0390, 0.9547]	[−0.0390, 0.9547]	$< 10^{-4}$
`power_mle` log-likelihood	−565.804	−565.804	$< 10^{-3}$
`groupwise_mle` $\hat{\beta}$	[1.0000, 1.9740, −0.9413]	[1.0000, 1.9740, −0.9413]	$< 10^{-4}$
`groupwise_mle` $\hat{\sigma}_g^2$ (groups A–D)	[0.274, 0.758, 1.950, 3.171]	[0.274, 0.758, 1.950, 3.171]	$< 10^{-3}$
`groupwise_mle` log-likelihood	−725.147	−725.147	$< 10^{-3}$

Values in both columns are rounded to 4 decimal places; the exact reference values and tolerances are in `test/runtests_r.jl`.

7.3 Historical note on test statistic formulas

An earlier HARE version used the generic LM form $\text{LM} = nR^2$ for both the Harvey and Glejser tests. This is asymptotically equivalent to the exact score-test formulas above but numerically different (e.g. 87.5 vs 122.0 for `harvey_test` on the het dataset), and it does not match `skedastic`. HARE was updated to use Harvey's (1976) and Glejser's (1969) exact formulas after the cross-validation revealed the discrepancy. The denominators $\pi^2/2$ and $\widehat{\sigma}^2(1-2/\pi)$ reflect the known variances of $\log \chi^2(1)$ and $|N(0,1)|$ respectively, making the score test exact rather than approximate.

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